

# 1 Solution — GIAM 7.1

## Problem 11

### 1.1 Problem

How many relations are there from a set of size  $n$  to a set of size  $m$ ?

### 1.2 Setup

Let  $A$  be a set with  $|A| = n$  and  $B$  a set with  $|B| = m$  (both  $n, m \geq 0$ ). Following the definition introduced in Chapter 6.1, a *relation from  $A$  to  $B$*  is, by definition, a subset of the Cartesian product:

$$R \subseteq A \times B.$$

So the set of all such relations is the power set  $\mathcal{P}(A \times B)$ , and we want

$$\# \text{relations} = |\mathcal{P}(A \times B)|.$$

### 1.3 Claim

$$\# \text{relations from } A \text{ to } B = 2^{nm}.$$

We give two proofs, both two-step.

### 1.4 Proof 1 — Reduction to Problems 8 and the Multiplication Principle

**Step 1: Count the underlying set.** By the multiplication principle, an ordered pair  $(a, b) \in A \times B$  is specified by  $a \in A$  ( $n$  choices) and  $b \in B$  ( $m$  choices), so

$$|A \times B| = n \cdot m = nm.$$

**Step 2: Count its subsets.** By Problem 8, any finite set with  $k$  elements has  $2^k$  subsets. Applying this with  $k = nm$ ,

$$|\mathcal{P}(A \times B)| = 2^{|A \times B|} = 2^{nm}. \quad \blacksquare$$

## 1.5 Proof 2 — Element-by-Element Decision

Building a relation  $R \subseteq A \times B$  from scratch amounts to making, for each pair  $(a, b) \in A \times B$ , an *independent* binary decision: “is  $(a, b) \in R$ , or not?” With  $nm$  pairs and 2 choices each, the multiplication principle gives  $2^{nm}$  relations.

Formally, the characteristic-function bijection

$$\chi : \mathcal{P}(A \times B) \longrightarrow \{0, 1\}^{A \times B}, \quad \chi(R)(a, b) = \begin{cases} 1 & \text{if } (a, b) \in R, \\ 0 & \text{otherwise,} \end{cases}$$

is the same one used in Problems 8 and 10. Its codomain has cardinality  $2^{|A \times B|} = 2^{nm}$  by Problem 10 (functions from a set of size  $nm$  into a set of size 2). Hence  $|\mathcal{P}(A \times B)| = 2^{nm}$ .  $\blacksquare$

## 1.6 Sanity Check on Small $(n, m)$

| $n$ | $m$ | $nm$ | $2^{nm}$ | Comment                                                                   |
|-----|-----|------|----------|---------------------------------------------------------------------------|
| 0   | any | 0    | 1        | $A \times B = \emptyset$ ; only the empty relation.                       |
| any | 0   | 0    | 1        | Same — Cartesian product with $\emptyset$ is $\emptyset$ .                |
| 1   | 1   | 1    | 2        | $A \times B = \{(a, b)\}$ ; two relations: $\emptyset$ and $\{(a, b)\}$ . |
| 2   | 2   | 4    | 16       | $A \times B$ has 4 pairs; $2^4 = 16$ subsets.                             |
| 2   | 3   | 6    | 64       | $A \times B$ has 6 pairs.                                                 |
| 3   | 3   | 9    | 512      | Already over 500 relations on a 3-element set!                            |

Table 1.1

The growth is doubly exponential in the dimensions: even modest  $n$  and  $m$  produce huge numbers.

## 1.7 Connection — Functions Are a Tiny Fraction of Relations

Compare with Problem 10 (number of functions  $A \rightarrow B$  is  $m^n$ ). A function is precisely a relation  $R \subseteq A \times B$  satisfying both of:

- **Existence.** For every  $a \in A$ , there is *some*  $b \in B$  with  $(a, b) \in R$ .
- **Uniqueness.** For every  $a \in A$ , this  $b$  is *unique*.

Most relations satisfy *neither*, which is why functions are so much rarer than relations:

| $(n, m)$ | Relations $2^{nm}$ | Functions $m^n$ | Ratio $\frac{2^{nm}}{m^n}$ |
|----------|--------------------|-----------------|----------------------------|
| $(2, 2)$ | 16                 | 4               | 4                          |
| $(2, 3)$ | 64                 | 9               | $\approx 7.1$              |
| $(3, 3)$ | 512                | 27              | $\approx 19.0$             |
| $(4, 4)$ | 65,536             | 256             | 256                        |

Table 1.2

The ratio  $2^{nm}/m^n$  grows rapidly. Concretely,  $f : A \rightarrow B$  corresponds to its *graph*

$$\Gamma_f = \{ (a, f(a)) : a \in A \} \subseteq A \times B,$$

and the map  $f \mapsto \Gamma_f$  realizes the inclusion  $\{\text{functions}\} \hookrightarrow \{\text{relations}\}$ .

## 1.8 Remark — The Special Case $A = B$

When  $A = B$ , the relations counted here are exactly the *relations on*  $A$  studied throughout Chapter 6. Taking  $n = m$  in our formula gives

$$\#\text{relations on an } n\text{-set} = 2^{n^2}.$$

Most of these have no special structure. The *named* subfamilies — equivalence relations, partial orders, functions, etc. — each impose constraints that knock the count down dramatically. For instance, on a **3**-element set there are  $2^9 = 512$  relations in total, but only **5** equivalence relations (the *Bell number*  $B_3 = 5$ ) and only **19** partial orders. The ambient  $2^{n^2}$  count is the “background” against which all of Chapter 6’s special relations live.